

Skill Extraction Engine

Day 9 Internship Deliverables

Submitted By

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Internship: AI Developer Intern

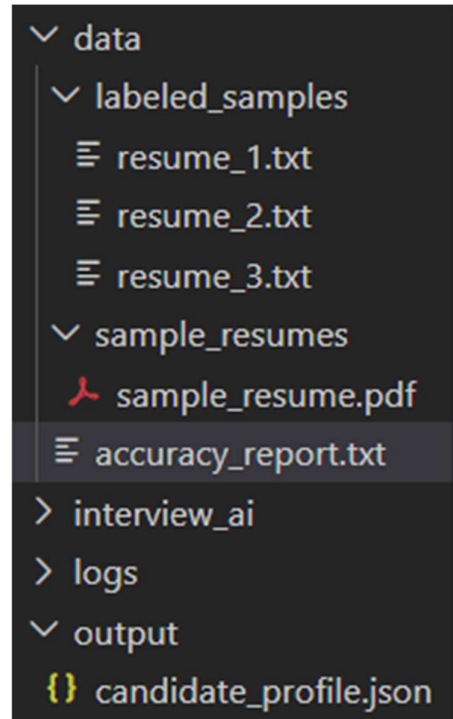
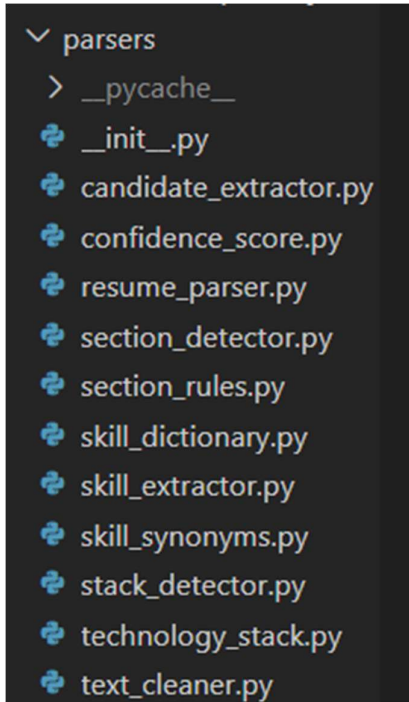
Date: 1 July 2026

Submitted To

Company: Zecpath

Department: Product & Engineering Team

1.Skill extraction engine



```
parsers > skill_extractor.py > SkillExtractor > extract_skills
1  import re
2
3  from parsers.skill_dictionary import (
4      TECHNICAL_SKILLS,
5      BUSINESS_SKILLS,
6      CREATIVE_SKILLS,
7  )
8
9  from parsers.skill_synonyms import SKILL_SYNONYMS
10
11
12  class SkillExtractor:
13      """
14      Extracts categorized skills from segmented resume sections.
15      """
16
17      def __init__(self):
18          self.technical = TECHNICAL_SKILLS
19          self.business = BUSINESS_SKILLS
20          self.creative = CREATIVE_SKILLS
21
22      def extract_skills(self, sections):
23          """
24          Extract skills from the segmented resume.
25          """
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + - [ ] [ ] ... [ ] [ ] X
```

```
PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System> python run_skill_extractor.py
```

```
Normalized Resume Text:
python, javascript, docker, machine learning worked with fastapi and postgresql built artificial intelligence and natural language processing applications
using tensorflow
```

```
Detected Skills
{'technical_skills': ['Artificial Intelligence', 'Docker', 'FastAPI', 'JavaScript', 'Machine Learning', 'Natural Language Processing', 'PostgreSQL', 'Python', 'TensorFlow'], 'business_skills': [], 'creative_skills': []}
```

3. Structured skill output

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell
PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System> python run_skill_extractor.py

Normalized Resume Text:
python, javascript, docker, machine learning worked with fastapi and postgresql built artificial intelligence and natural language processing using tensorflow

Detected Skills
{'technical_skills': ['Artificial Intelligence', 'Docker', 'FastAPI', 'JavaScript', 'Machine Learning', 'Natural Language Processing', 'TensorFlow'], 'business_skills': [], 'creative_skills': []}

Technology Stack
{'Frontend': ['JavaScript'], 'Backend': ['Python', 'FastAPI'], 'Database': ['PostgreSQL'], 'AI/ML': ['Artificial Intelligence', 'Machine Learning', 'Natural Language Processing', 'TensorFlow'], 'Cloud & DevOps': ['Docker']}

Skill Confidence Scores
{'skill': 'Artificial Intelligence', 'confidence': 0.97}
{'skill': 'Docker', 'confidence': 0.99}
{'skill': 'FastAPI', 'confidence': 0.99}
{'skill': 'JavaScript', 'confidence': 0.99}
{'skill': 'Machine Learning', 'confidence': 0.97}
{'skill': 'Natural Language Processing', 'confidence': 0.97}
{'skill': 'PostgreSQL', 'confidence': 0.99}
{'skill': 'Python', 'confidence': 0.99}
{'skill': 'TensorFlow', 'confidence': 0.99}
PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System>
```

